

Hybrid Search & Retrieval POC

https://github.com/aiyshwarya/Search_Engine

Python

FAISS

BM25

OpenSearch

Sentence Transformers

Tesseract

OVERVIEW

A production-style multilingual hybrid search engine for Indian-language documents, combining BM25 lexical search and FAISS semantic retrieval to rank documents with snippet-level evidence across Tamil, Hindi, Telugu and more.

IMPACT

Built a multilingual hybrid search pipeline supporting Indian-language PDFs and DOCX files, combining BM25 (OpenSearch) and FAISS IVF-PQ for accurate document retrieval with chunk-level evidence and best-snippet selection.

TECHNICAL WALKTHROUGH

Elevator pitch

This repo is a small production-style hybrid search pipeline for PDFs/

DOCX: it ingests files, extracts text (OCR fallback), deduplicates, token-chunks documents, precomputes multilingual embeddings, builds FAISS ANN indexes, and serves a hybrid search that combines BM25 (OpenSearch) + ANN (FAISS) to rank whole documents with snippet-level evidence.

One-minute explanation (what to say first in an interview)

“We built a hybrid retrieval pipeline where BM25 provides precise lexical matches and FAISS (IVF+PQ + optional HNSW hot tier) provides semantic recall. Documents are normalized, chunked to ~400–800 tokens, embedded with a multilingual MiniLM, then indexed. At query time we run BM25 and ANN in parallel, union chunk candidates, aggregate per-doc signals (max BM25, max vector), and return ranked documents with a snippet and page offset.”

System components — 30-second walk-through

Ingest (optimized_ingest.py): PDF/DOCX extraction, Tesseract OCR

fallback, per-page OCR confidence, language detection (fastText + Lingua).

Dedup (dedupe_simhash.py): document-level Simhash dedupe to avoid redundant indexing.

Chunking (chunker.py): tokenizer-based chunking with overlap (target tokens ~512, overlap ~64).

Embeddings (embedder.py): batch embed with sentence-transformers; normalize embeddings and write .npy + meta.

ANN build (build_faiss.py): create IVF+PQ (and HNSW for hot subsets) from embeddings.

Search (search_service.py): OpenSearch BM25 + FAISS ANN in parallel, union & aggregate to score docs.

KEY DETAILS

- Ingests PDF and DOCX files with digital text extraction and Tesseract OCR fallback for scanned documents.
- Detects document language using FastText (LID176) across Tamil, Hindi, Telugu, Kannada, Bengali and more.
- Applies SimHash near-duplicate detection and token-aware chunking with sentence-based splitting.
- Generates multilingual embeddings using Sentence Transformers and indexes them in FAISS IVF-PQ.
- Hybrid BM25 and ANN retrieval with document-level score aggregation and best-snippet selection per result.